

# Overnight progress

## EMBER

Real inputs. GPU execution. Fire modelling.

Team progress update · 29 September 2026

# Real terrain and wind

## Elevation

Real heights aligned with the Collserola simulation grid

## Wind

ERA5 reanalysis snapshot for a selected date and hour

## Richer inputs for reproducible test cases

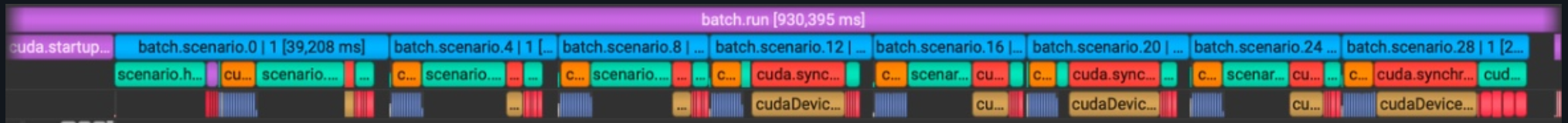
Current route: scalar CPU • Uniform, constant wind per run

The simulation is not yet a calibrated wildfire forecast

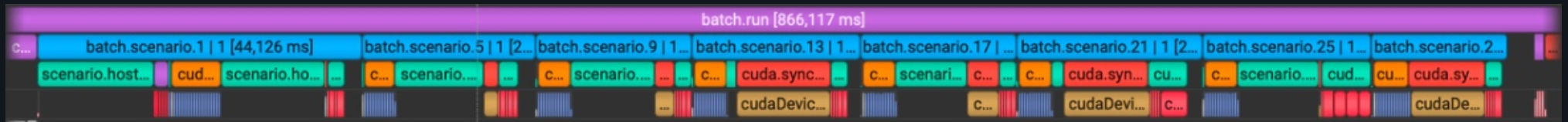
# GPU execution profiling

Julián / Profiling scenario execution across four MPI processes

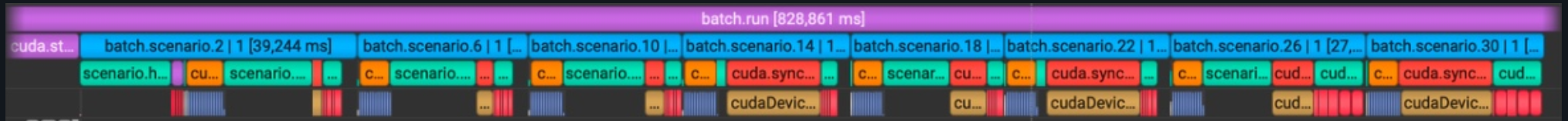
MPI 0



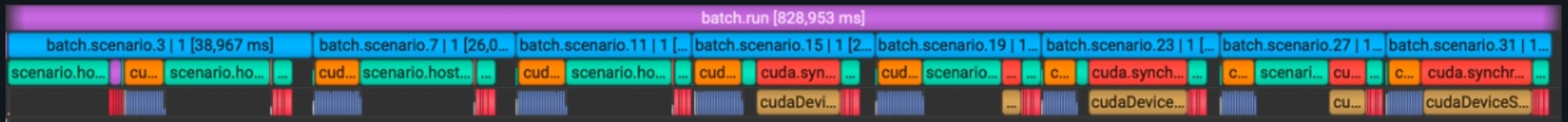
MPI 1



MPI 2



MPI 3



Execution traces guide optimisation; speedup still needs a matched benchmark

# Fire model development

Jesús / Two approaches investigated in issue #4

## rothermel-fire-model

Implemented • Compilation and testing pending

## normalized-fuel-model

Branch created for the alternative approach

Both branches created from dev

# Next steps

- 01 Compile and test Rothermel
- 02 Benchmark the GPU changes
- 03 Work towards integration

Proposed next steps • Integration and validation remain ahead